

Navigators Technical Documentation

Navigators — Experimental Methodology & Ablation Protocol

Document Identifier: EXP-NAV-01
Status: Production Reference
Version: 1.0.0
Last Updated: 2026-09-23
Attribution: Developed by Navigators

1. Experimental Design Philosophy

Navigators adheres to strict empirical guidelines for navigation benchmarking:

- 1. **Zero Ground-Truth Leakage:** During simulated or real GNSS outage intervals, GNSS measurements (including derived GNSS speed and ground-track heading) are completely withheld from the estimator. GNSS coordinates are sequestered strictly as an evaluation reference to compute positional error post-run.
- 2. **Identical Inputs Across Ablations:** To isolate the exact marginal utility of each algorithmic component, ablations A through G are evaluated on the exact same sensor stream, calibration prefix, and outage window.
- 3. **Deterministic Reproducibility:** Random seeds, filter tuning matrices, normalization parameters, and input dataset hashes are recorded in test logs.

2. Seven-Stage Ablation Hierarchy (A through G)

The evaluation suite (`replay.py` and `scripts/benchmark.py`) defines seven progressive system configurations:

Mode	Identifier	Filtering	AI Velocity	15-State EKF	NHC Constraints	ZUPT Detector	Map Matching
A	Raw IMU Baseline	✗	✗	✗	✗	✗	✗
B	Filtered IMU	✓	✗	✗	✗	✗	✗
C	AI Velocity Only	✓	✓	✗	✗	✗	✗
D	AI + EKF Fusion	✓	✓	✓	✗	✗	✗
E	AI + EKF + NHC	✓	✓	✓	✓	✗	✗
F	AI + EKF + NHC + ZUPT	✓	✓	✓	✓	✓	✗
G	Full Production System	✓	✓	✓	✓	✓	✓

Architectural Roles of Each Stage:

- **Mode A (Raw IMU):** Direct double-integration of aligned raw accelerometer readings. Illustrates unconstrained MEMS drift where quadratic position error accumulates rapidly (> 500 m in 60s).
- **Mode B (Filtered IMU):** Incorporates running median and causal 20 Hz low-pass filtering. Suppresses engine acoustic noise spikes.
- **Mode C (AI Velocity Direct):** Directly integrates TCN-predicted 2D velocity vectors without Kalman filtering: $\mathbf{p}_k = \mathbf{p}_{k-1} + \mathbf{v}_{tcn}\Delta t$.
- **Mode D (AI + EKF):** Fuses TCN velocity as a pseudo-measurement into the 15-state EKF, actively estimating accelerometer and gyroscope biases.
- **Mode E (AI + EKF + NHC):** Activates Non-Holonomic Constraints ($v_y^V \approx 0, v_z^V \approx 0$), eliminating lateral vehicle slide and vertical altitude drift.
- **Mode F (AI + EKF + NHC + ZUPT):** Integrates zero-velocity updates during traffic stops, freezing position error accumulation during stationary intervals.
- **Mode G (Full System):** Snaps the filtered state trajectory to OpenStreetMap road centerlines, eliminating cumulative cross-track angular drift.

3. Evaluation Metrics & Mathematical Definitions

All positioning metrics are computed in horizontal meters relative to the local East-North-Up (ENU) tangent origin:

3.1 Absolute Trajectory Error (ATE RMSE)

Root Mean Squared Error between estimated coordinates $[p_E, p_N]$ and ground-truth reference coordinates $[p_E^{ref}, p_N^{ref}]$ over scorable timesteps N :

$$\text{ATE RMSE} = \sqrt{\frac{1}{N} \sum_{k=1}^N \left((p_{E,k} - p_{E,k}^{\text{ref}})^2 + (p_{N,k} - p_{N,k}^{\text{ref}})^2 \right)}$$

3.2 Maximum & Final Position Drift

- **Maximum Drift:** $\max_k \| \mathbf{p}_k - \mathbf{p}_k^{\text{ref}} \|$ over the outage duration.
- **Final Position Drift:** Euclidean error at the concluding epoch of the outage window:

$$e_{\text{final}} = \| \mathbf{p}_{\text{outage_end}} - \mathbf{p}_{\text{outage_end}}^{\text{ref}} \|$$

3.3 Velocity Vector RMSE

$$\text{Velocity RMSE} = \sqrt{\frac{1}{N} \sum_{k=1}^N \left((v_{E,k} - v_{E,k}^{\text{ref}})^2 + (v_{N,k} - v_{N,k}^{\text{ref}})^2 \right)}$$

3.4 Heading Angular Error (RMSE)

Wrapped angular error in degrees, computed strictly when vehicle ground speed exceeds 0.5 m/s to avoid stationary gimbal noise:

$$\Delta\psi_k = \text{wrap}_{180}(\psi_k - \psi_k^{\text{ref}})$$

$$\text{Heading RMSE} = \sqrt{\frac{1}{M} \sum_{j=1}^M (\Delta\psi_j)^2}$$

4. Execution Protocols

4.1 Running the Benchmark Suite

To execute the automated ablation suite against an input dataset:

```
# Replay full seven-stage ablation on a trip recording:
python replay.py --dataset data/raw_trips/held_out_trip.json --gnss-outage 60 --outage-start 10
--ablations

# Equivalent benchmark script:
python scripts/benchmark.py --dataset data/raw_trips/held_out_trip.json --gnss-outage 60
```

4.2 Output File Artifacts

The evaluation writes structured audit artifacts to `results/replay/` : - `trip_results.json` : Full configuration parameters, component call counts, execution timings, and numerical metrics across whole-trip, outage, and recovery intervals. - `trip_[A-G]_trajectory.csv` : Detailed timestamped state trajectories ($t, p_E, p_N, v_E, v_N, \psi$, mode, GNSS allowed). - `trip_[A-G]_trajectory.json` : Interactive playback files renderable directly in the browser dashboard. - `trip_ablation.csv` : Consolidated side-by-side comparison table of all modes.

Developed by Navigators